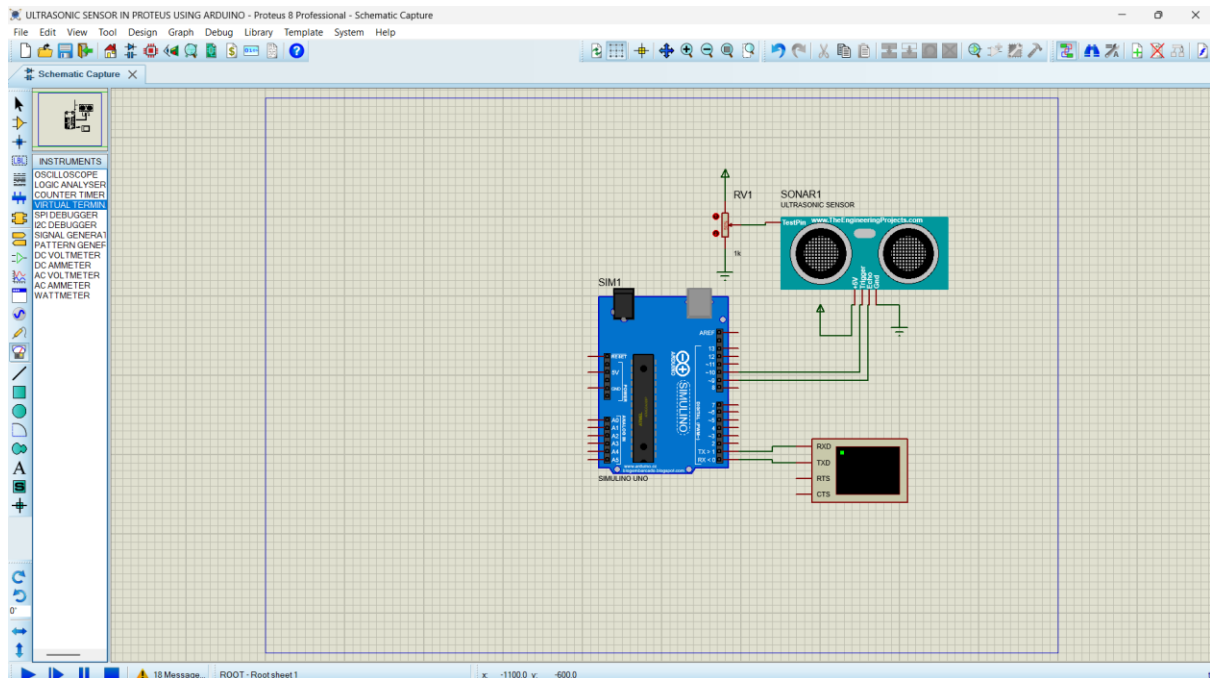


Exp-9: ULTRASONIC SENSOR IN PROTEUS USING ARDUINO

1) Before Running the code:



2) Source Code

// Program Sketch

```
int trig = 10;
```

```
int echo = 9;
```

```
long duration;
```

```
int cm;
```

```
void setup() {
```

```
// put your setup code here, to run once:
```

```
pinMode(trig, OUTPUT);
```

```
pinMode(echo,INPUT);
```

```
Serial.begin(9600);
```

```
}
```

```
void loop() { // Sketch to calculate ultrasonic distance
```

```
digitalWrite(trig, LOW);
```

```
delayMicroseconds(10);
```

```

digitalWrite(trig, HIGH);

delayMicroseconds(10);

digitalWrite(trig, LOW);

delayMicroseconds(10);

duration = pulseIn(echo, HIGH);

cm = (duration/2) * 0.034;

// Print Value on Serial Monitor

Serial.print(" Distance = ");

Serial.print(cm);

Serial.println(" cm");

}

```

3) After Simulation

